Chapter Six

Future Work

6.1 Introduction

This chapter presents the current status of the Smart Basket Market — ScanGo project, highlighting the progress achieved during the current semester and identifying the remaining tasks for the upcoming development phase. It also outlines the necessary steps required to transform the system from the design and prototyping phase into a fully integrated and deployable smart retail solution.

The project focuses on automating the customer journey within a retail environment through a smart shopping basket system. Rather than automating the entire store, the system targets only the customer-facing experience, aiming to improve the shopping process by reducing reliance on cashiers, minimizing waiting time, and eliminating congestion at checkout points.

6.2 Completed Work

During the current phase, several key achievements have been accomplished, forming the foundation for future development. These components collectively establish the architectural and design basis upon which the full system will be built.

6.2.1 Problem Analysis and Concept Definition

The inefficiencies of traditional checkout systems were analyzed, with particular focus on long waiting times and congestion resulting from full dependence on cashiers. Based on this analysis, the Smart Basket Market concept was proposed, enabling customers to scan products independently as they place them in the basket, thereby streamlining the entire checkout process without requiring dedicated cashier intervention.

6.2.2 System Requirements Analysis

Functional and non-functional requirements were defined in Chapter 3 and are briefly referenced here to maintain consistency throughout the system design. These requirements serve as the governing constraints for all subsequent development decisions.

6.2.3 System Design

An initial system design was developed to clarify user interactions and define the core components of the system. The design artifacts produced include:

- Workflow design illustrating the full customer journey
- Use Case Diagrams defining system actors and interactions
- Sequence Diagrams depicting runtime communication between components
- Class Diagrams representing the system data model

6.2.4 UI/UX Design

The user interfaces for the smart basket tablet application were designed with a focus on simplicity and accessibility. The design prioritizes a user-friendly experience that allows any customer to interact with the system without prior training, while maintaining a smooth and responsive workflow throughout the shopping session. The integrated payment module within the tablet application was also designed to ensure a seamless and intuitive checkout experience directly from the basket device.

6.3 Technical Implementation (Upcoming Phase)

The next phase will focus on transforming the system design into a fully functional solution through a structured development approach that covers the complete customer journey — from entering the store to exiting through the smart gate.

6.3.1 Smart Basket Application Development (Flutter)

The smart basket application will be developed using Flutter and installed on a low-cost Android tablet or recycled smartphone attached to each basket. The application serves as the primary point of interaction for the customer during the shopping session, and incorporates the payment functionality directly within the same interface. Key features include:

- Barcode scanning using the device camera to identify products
- Adding and removing products from the active basket session
- Updating product quantities in real time
- Displaying the live invoice as items are scanned
- Generating a QR code representing the finalized invoice upon checkout
- Reviewing the complete invoice details before confirming payment
- Selecting a preferred payment method

- Completing a secure electronic payment transaction directly on the basket tablet

A weight sensor integrated into the basket will serve as an additional verification layer, ensuring that scanned items correspond to the actual contents of the basket. This mechanism helps detect unscanned items and supports system integrity.

The payment module embedded within the basket application is designed to be scalable, with the architecture supporting the integration of additional payment methods in future iterations.

6.3.2 Database Development

A relational database using SQL Server will be designed with indexed tables to ensure efficient query performance and fast retrieval of product and transaction data. The database will store:

- Product data including name, price, and barcode identifier
- User account data and authentication records
- Transaction records and payment history

The database design will prioritize fast access, high data integrity, and efficient management to support real-time system operations across multiple concurrent sessions.

6.3.3 Model and Algorithm Integration

The system will integrate barcode recognition algorithms to ensure accurate and fast product identification during the scanning process. In addition, a basic recommendation system based on user purchase history will be incorporated to enhance the personalized shopping experience. The recommendation approach will follow a content-based filtering technique, suggesting similar products based on user preferences and previously scanned items. These intelligent components will further enhance system value and provide dashboards for visualizing aggregated results and user insights.

6.3.4 Technology Stack Justification

The system technologies were carefully selected to ensure performance, scalability, and ease of development across all components:

- Flutter is used for cross-platform development, enabling the application to run efficiently on both customer mobile devices and smart basket tablets, including the integrated payment module
- SQL Server is selected for its scalability, reliability, and ability to handle structured data efficiently

- ASP.NET is used to develop robust and secure backend services that support system integration and API management
- ESP8266 is utilized for low-cost and efficient IoT integration, enabling real-time communication with the smart gate hardware

6.4 System and Hardware Integration

This phase is critical as it transforms the individually developed components into a unified, cohesive system capable of supporting the full customer journey.

6.4.1 System Integration

All system components will be connected, including linking the basket application to the central database, integrating the payment module with the invoice engine, and ensuring real-time data synchronization across all active sessions.

6.4.2 Backend Integration

The backend built with ASP.NET will be integrated with the basket tablet application — which now incorporates the payment functionality — to support the full customer journey, beginning from session initiation upon entering the store and concluding with payment confirmation at the exit gate.

6.4.3 Hardware Integration (Smart Gate)

The system will be integrated with an IoT-based gate control mechanism using the ESP8266 module to ensure secure exit verification. Upon completing the shopping session, the customer will perform a secondary scan of the purchased items at the exit gate. The smart gate will verify that the scanned items match the paid invoice before granting exit clearance, providing a reliable loss-prevention mechanism without requiring cashier involvement.

6.5 Security and Verification

A comprehensive security mechanism will be implemented to ensure safe transactions and prevent misuse of the system. The security framework encompasses the following components:

- Encryption of all payment data during transmission and storage
- A verification system at the exit gate that cross-checks scanned items against the confirmed payment record

- Real-time payment status checking at the gate (Paid / Unpaid) to prevent unauthorized exit

6.6 Risk Analysis

The following risks have been identified as potential threats to system reliability and continuity:

- Network failure during active shopping sessions or payment transactions
- Hardware malfunction in the smart basket tablet, weight sensor, or gate system
- Payment processing failure due to gateway or connectivity issues

The following mitigation strategies have been defined to address these risks:

- Offline caching for temporary transaction and session data storage during network interruptions
- A backup verification mechanism at the exit gate to handle connectivity failures
- Retry mechanisms for failed payment transactions to ensure transaction completion

6.7 Deployment Plan

The system will be deployed using cloud-based infrastructure, such as Microsoft Azure or AWS, to ensure scalability, high availability, and geographic redundancy. Backend services will be containerized using Docker to simplify deployment, maintenance, and horizontal scaling. APIs will be hosted in a secure cloud environment to support seamless communication between the basket application and database.

6.8 Testing and Evaluation

The system will undergo comprehensive testing across multiple dimensions to ensure correctness, performance, and security before deployment:

- Functionality Testing: ensuring accurate barcode scanning, correct invoice generation, and reliable QR code transactions
- Performance Testing: evaluating system response speed, concurrent session handling, and overall stability under load
- Security Testing: verifying data protection mechanisms and preventing unauthorized system access

- User Experience Testing: validating ease of use for customers unfamiliar with the system

Real-world retail scenarios will also be simulated to validate end-to-end system performance under conditions representative of actual deployment environments.

6.9 Proposed Implementation Timeline

The project will be implemented across four structured phases as outlined in the following table:

Phase	Description	Duration
Phase 1	Flutter application development (Smart Basket with integrated Payment)	3 weeks
Phase 2	QR invoice generation, integrated payment module, and database development	3 weeks
Phase 3	Full system integration, IoT smart gate, and hardware setup	2 weeks
Phase 4	Testing, optimization, and final deployment	2 weeks

6.10 Summary

This chapter outlined both the achievements completed during the current semester and the detailed roadmap for future development. By combining advanced software technologies with IoT hardware integration, the Smart Basket Market — ScanGo system aims to deliver a smart, efficient, and scalable shopping experience suitable for real-world retail environments.

The system's key innovation lies in automating only the customer journey: customers scan products into their smart baskets, receive a QR-coded invoice on the tablet, complete payment independently through the integrated payment module on the same basket tablet, and exit through the smart gate after a secondary verification scan. This approach eliminates traditional checkout bottlenecks while maintaining security and data integrity throughout the process. The success of the upcoming phase depends on proper implementation, rigorous integration testing, and thorough evaluation to ensure reliability and performance at scale.